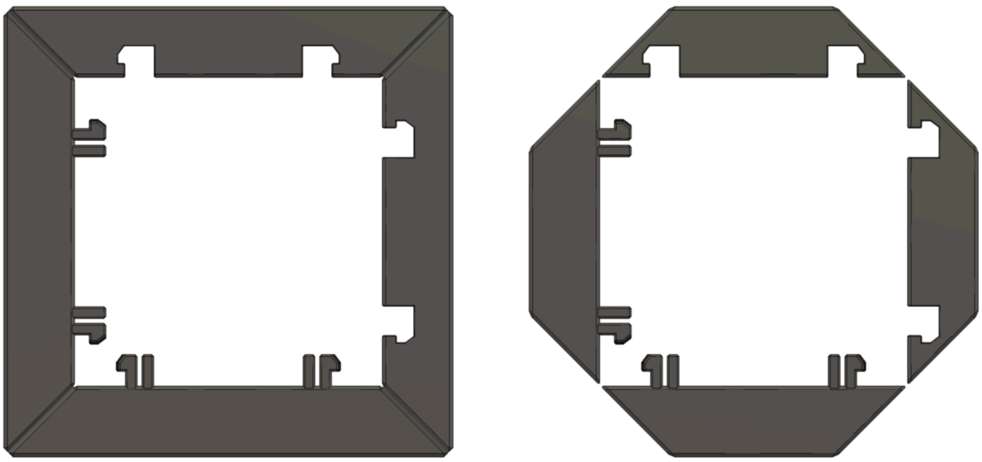
PLAY CONVEYOR

CARGO CONNECT: FRAMES



User Guide

Resources, Print Settings and Assembly

Before You Start..

What's Included

The below files and documents are included within your download:

1. **Ready to Print (.3mf file) -** All the parts in one place and ready to print on Bambu Lab printers.
2. **User Guide (.pdf file) -** A complete 'User Guide' to help you print and assemble the model.
3. **Individual parts (.stl files) -** All the 3D printable parts you need to make the complete base model, compatible with most slicers.
4. **Assembly View Only file (Do Not Print) -** This file is to view the fully assembled model only and not for printing.

What We Used

Printers:

Bambu Lab P1S

You can get the printers we used here (affiliate link)* **- Bambu Lab P1S**

Filaments:

Cargo Connect Frames - Bambu Lab PLA Basic Dark Grey

You can get the filaments we used here (affiliate link)* **- Bambu Lab Filament**

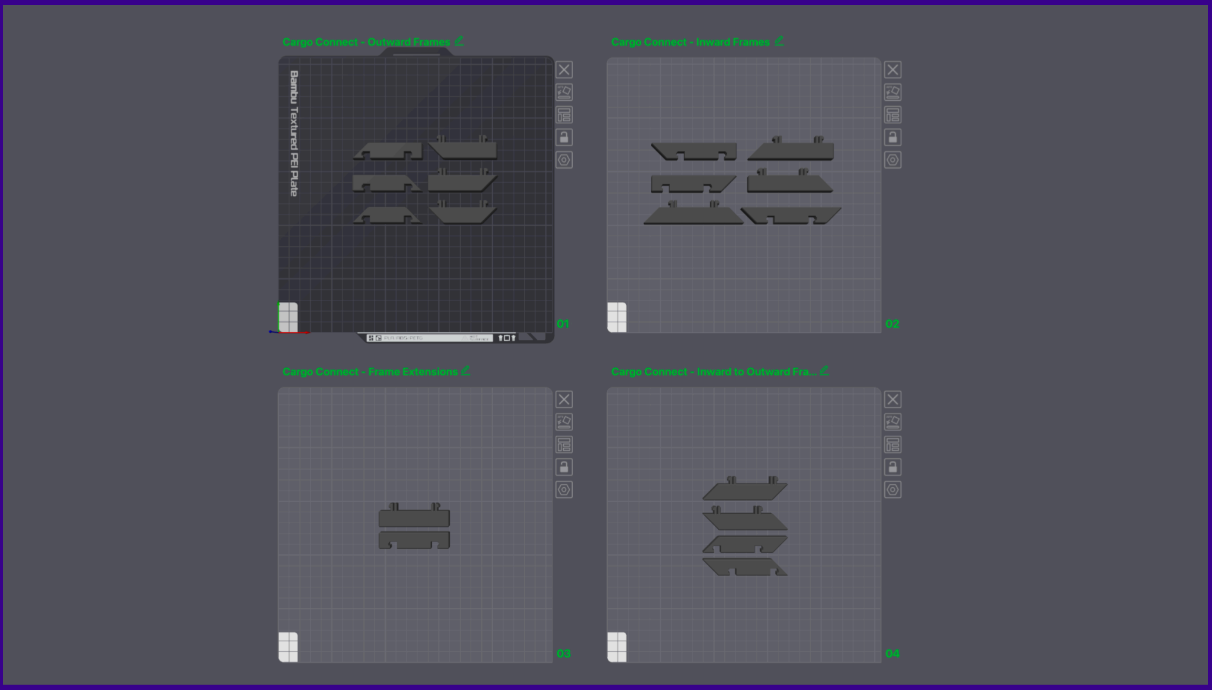
*Some links in this document are affiliate links, which means we may earn commission if you purchase through them.

Print Settings

Important Note:
The print settings listed here are the ones that worked best for us and for our specific needs, using our own printers, materials, and environment. However, every setup is different - so you may need to adjust these settings to suit your printer, filament, and your specific needs.

Ready to Print Project

A .3mf file named **'1.Ready to Print'** is included in the download. This file is a pre-prepared Bambu Studio project which includes the settings we used. The build plates are organised into groups of Frames based on their orientation — you can simply copy and paste the frames you need more of, and delete any you don't need at this time. (Refer to 'Assembly Guide - Step 1' for more information on selecting the frames to print)



Individual Part Settings

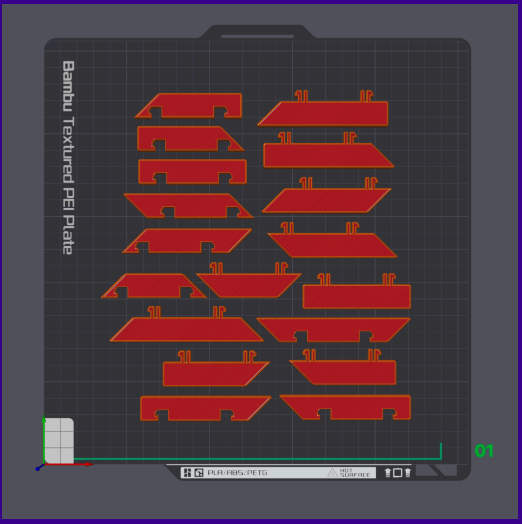
Frames

Basic Settings:

- **Printer** - Bambu Lab P1S
- **Outer Walls** - 2
- **Infill** - 15%
- **Supports** - None
- **Brim** - Auto

Estimations:

- **Print Time** - <10m per frame
- **Filament Used** - <7g per frame

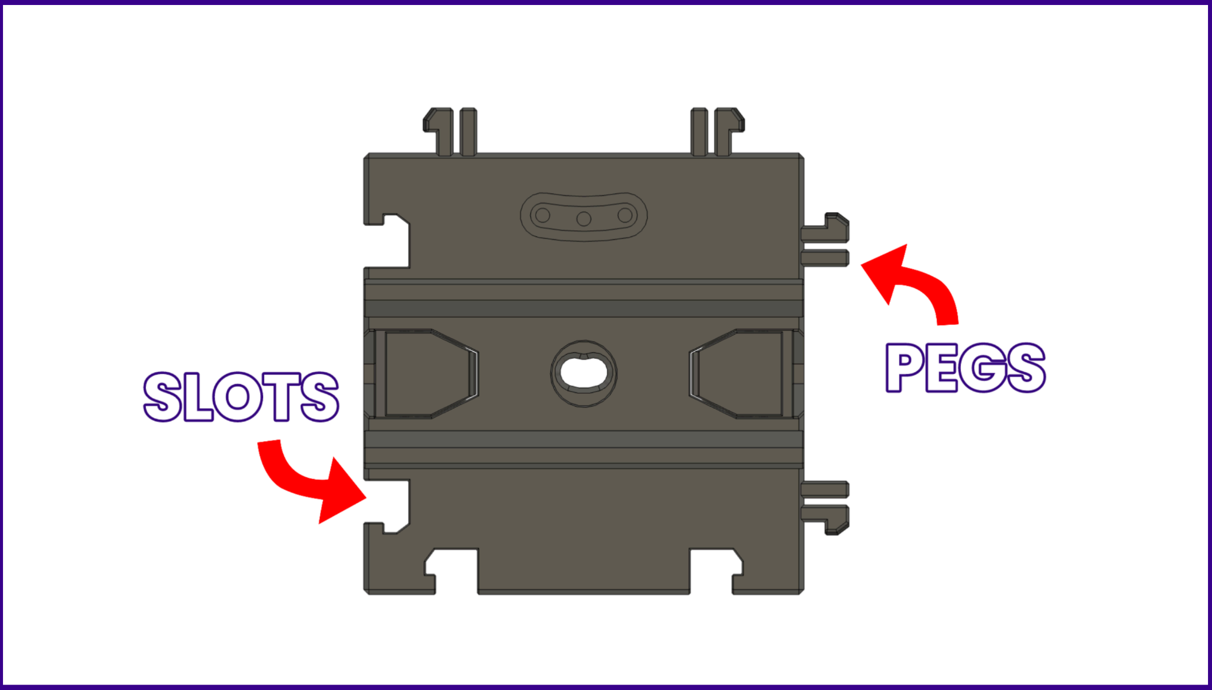


Assembly Guide

Important Note:
Once all parts are printed, remove any supports and check for any imperfections or poorly printed areas - *these may need post processing work or reprinting.*

Step 1 - Selecting your Frames

The Cargo Connect Interlocking System is designed to click together infinitely, which means each tile has a specific orientation: **pegs on the top and right**, and **slots on the bottom and left,** as shown below.



To make the frames compatible with every possible edge and corner, we've included different versions for each angle and tile side. You'll just need to match the correct frame type to the position and orientation of the tile you're framing.

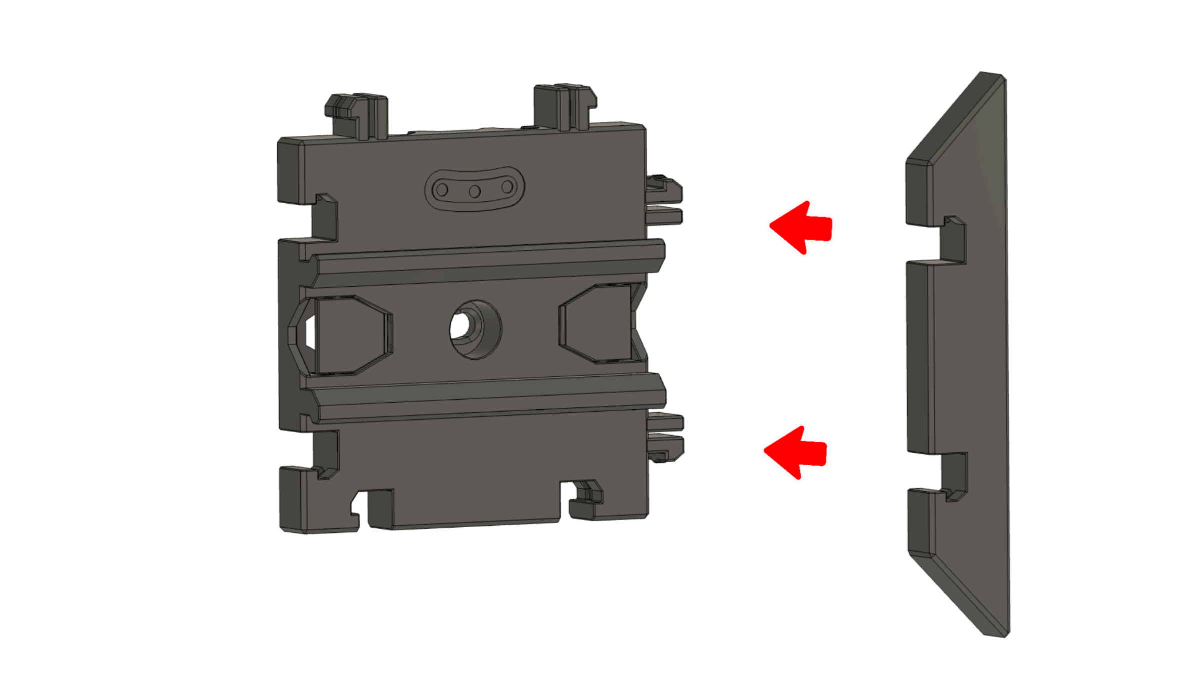
The frames are grouped into four categories:

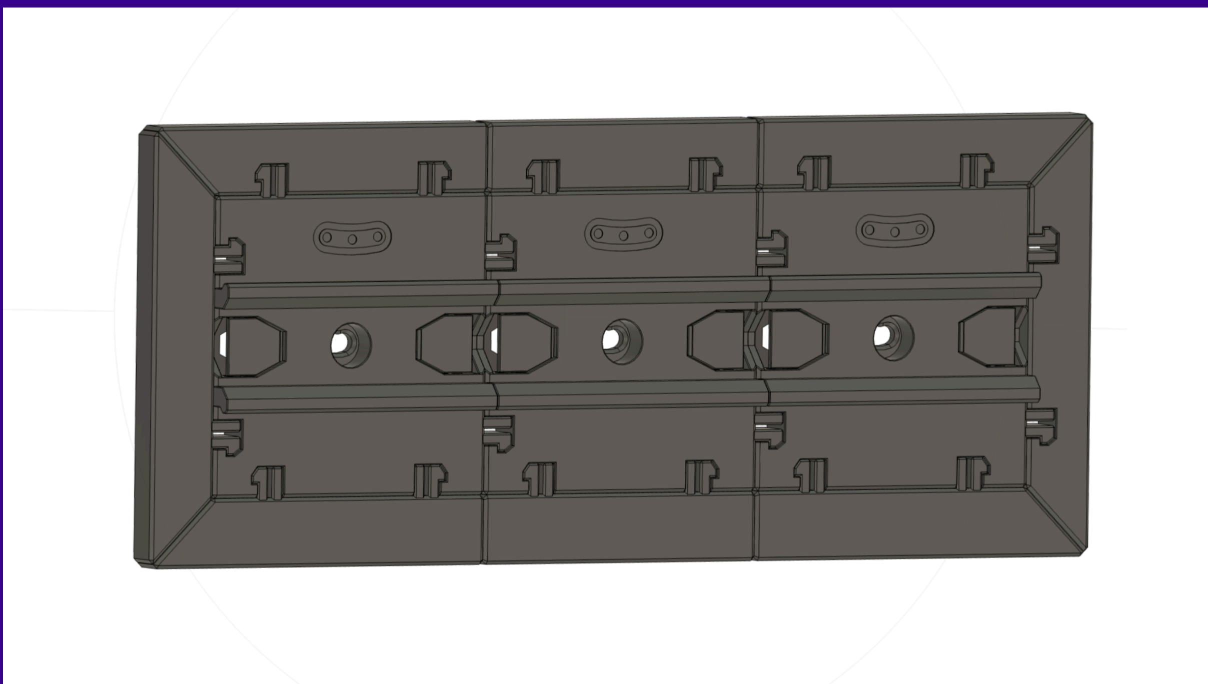
- **Outward Frames** – frames with 45° corners that point *away* from the tile
- **Inward Frames** – frames with 45° corners that point *toward* the tile
- **Inward to Outward Frames** – frames with corners that angle in opposite directions
- **Extension Frames** – flat edge pieces used for clean, straight runs between corners

Each group includes versions for **pegs** and **slots**, depending on which side of the tile the frame will connect to. Simply choose the STL that matches your layout and tile orientation, and you're good to go.

Step 2 - Use

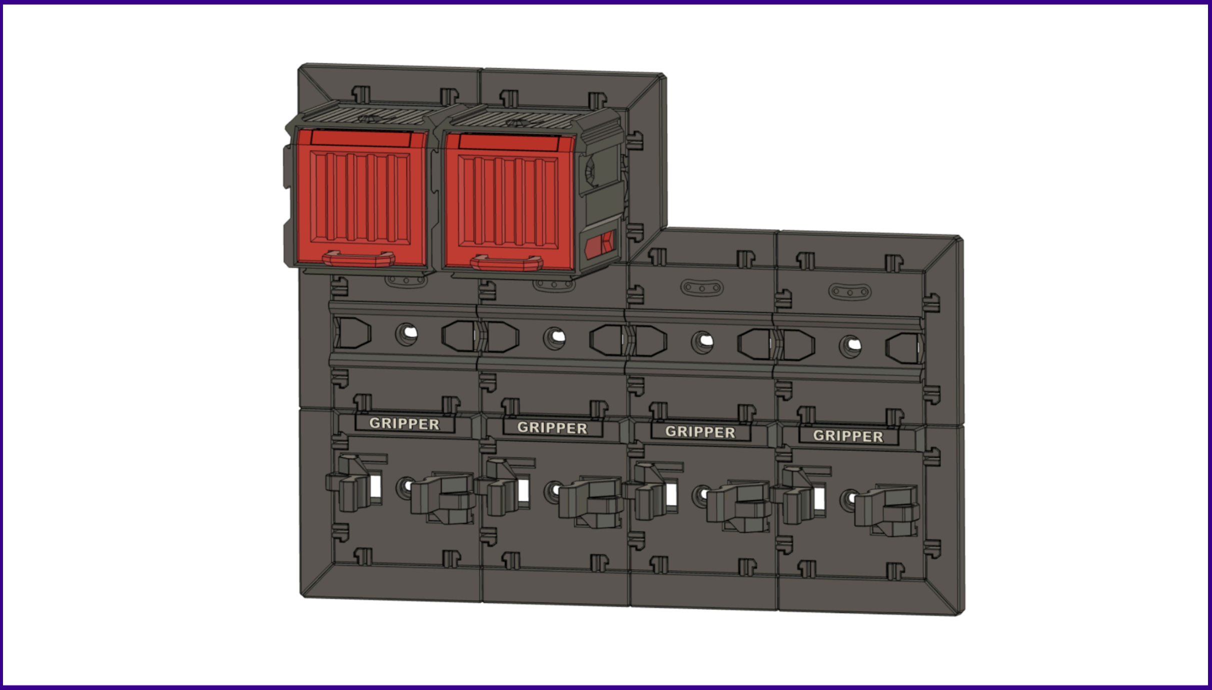
Once you have the Frames you need for your wall setup, they simply click into any Cargo Connect tile using the Interlocking System around the outside, as shown below:





Compatible Components

The Cargo Connect: Frames can interlock with all Cargo Connect tiles, meaning you can build the perfect storage system for all your needs.



Troubleshooting

We put a lot of time into making our designs as easy to print and use as possible, testing extensively with various filaments and printers. That said, we understand that occasional issues can occur - below are some potential solutions that may be useful:

Warping and Bed Adhesion

Warping is a common challenge in FDM 3D printing, often caused by poor bed adhesion, rapid cooling, or inconsistent ambient temperatures.

It's important that tiles and components come off the bed as flat as possible, as warping may impact features like interlocking pegs, sliding fits, or hinge alignment. If you notice lifting corners or uneven bases, try adding a brim or applying a suitable adhesive directly to the build plate to improve bed adhesion.

Ensuring consistent ambient temperatures and proper first-layer calibration can also go a long way in reducing warping across prints.

Tolerances

The Cargo Modular System is designed to be as print-friendly as possible, and we've worked hard to make tolerances as forgiving as possible across a wide range of printers. However, the unique features and functionality of our designs can rely on tight tolerances to work as intended — especially for parts that slide, hinge, or interlock.

If you're experiencing issues with fit or movement, it's worth checking your printer calibration — including flow rate, retraction settings, and nozzle temperature — to ensure accurate extrusion. Also make sure your filament is dry and of consistent quality, and that your slicer profile is correctly set up for your printer and material.

If problems still persist, we recommend printing a tolerance test model to help identify where your setup might need fine-tuning. These are widely available from reputable sources such as Thangs.

Use Disclaimer

Every Play Conveyor model has been designed and tested with care, but results will vary depending on your printer, materials, settings, and environment. Because of this, we cannot guarantee print outcomes across all setups.

You are responsible for ensuring that both the printing and assembly of any models are completed to a safe and functional standard. Always test your prints — especially components that are wall-mounted, load-bearing, or move — and check them regularly for wear or failure.

This user guide is provided as a general reference to support your build, but it is not exhaustive or guaranteed to suit all applications. Use your judgment where needed to suit your specific setup.

All files are provided "as is" and without warranty. Play Conveyor Ltd accepts no responsibility for any injury, damage, or loss resulting from the use or misuse of our designs.

This disclaimer highlights some of the key points when using our models, but it does not replace or override the official 'License Agreement'.

The 'License Agreement' associated with your subscription tier is the official document that defines your permitted use of the model, any restrictions, and your responsibilities. Please ensure you review the relevant license agreement to confirm proper usage.

Thangs Members

- Maker tier: covered by the **Personal License** (for personal use only)
- Merchant tier: covered by the **Commercial License** (includes permitted commercial use)

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